HK IMO Training 2023-2024
Problem Set 3 Suggested Solutions

Problem 9 (Durer Math Competition Finals 2019 Category E Problem 3)**.** Find all triples (p, q, r) of prime numbers such that $p^q + p^r$ is a perfect square.

Solution. Answer: $(p, q, r) = (2, 2, 5), (2, 5, 2), (3, 2, 3), (3, 3, 2), (2, q, q)$ where q is any odd prime.

WLOG assume $q \geqslant r$. If $q = r$, then we need $2p^q$ to be a perfect square. Since the exponent of 2 should be even, we must have $p = 2$. Then $2p^q = 2^{q+1}$ is a perfect square if and only if q is an odd prime.

If $q > r$, we have $p^q + p^r = p^r(p^{q-r} + 1)$. Since $p \nmid p^{q-r} + 1$ and the exponent of p should be even, r must be even. As r is a prime, we must have $r = 2$. Now, we need $p^{q-2} + 1 = k^2$ for some positive integer k. This implies

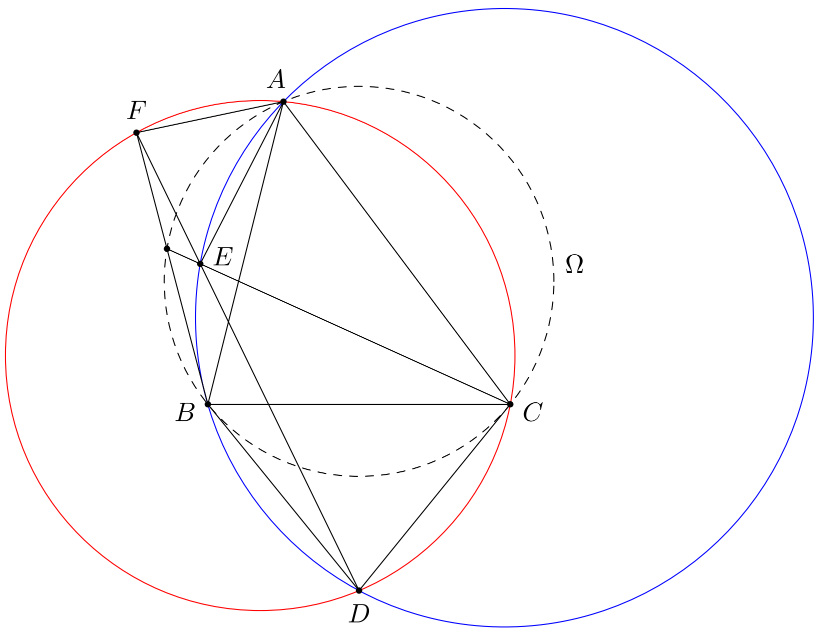
$$p^{q-2} = k^2 - 1 = (k-1)(k+1).$$

Then each of $k - 1$ and $k + 1$ is a power of p. Note that $(k - 1, k + 1) = (k - 1, 2)$ is equal to 1 or 2.

- If $k - 1 = 1$, we have $k = 2$. Then $p^{q-2} = 3$, and hence $p = 3, q = 3$.
- If $k - 1 > 1$, we have $p = 2$ as both $k - 1$ and $k + 1$ are powers of p. Also, as $k - 1$ and $k + 1$ differ by 2, and both are powers of 2, we must have $k - 1 = 2$ and $k + 1 = 4$. This means $k = 3$. Then $p^{q-2} = 8$, and hence $p = 2, q = 5$.

$\square$

Problem 10 (Caucasus MO 2022 Senior Problem 4)**.** In acute $\triangle ABC$, the tangents at B and C to the circumcircle Ω of $\triangle ABC$ meet at D. A line passing through D cuts the circumcircles of $\triangle ABD$ and $\triangle ACD$ again at E and F respectively. Prove that the lines CE and BF intersect on Ω.



Solution. Using directed angles, we have

$$\measuredangle AFE = \measuredangle AFD = \measuredangle ACD = \measuredangle ABC$$

and similarly $\measuredangle AEF = \measuredangle ACB$. This shows $\triangle AFE$ and $\triangle ABC$ are directly similar. Thus, A is the centre of spiral similarity mapping FE to BC. It is well-known that the intersection point of FB and EC lie on $(ABC) = \Omega$. $\square$

Problem 11 (Caucasus MO 2022 Senior Problem 7)**.** We first choose a positive integer n, then write down the polynomial $(x+1)^n$ in its expanded form, and change every coefficient to its reciprocal. For example, if we choose $n=3$, then the resultant polynomial is $x^3+\frac{1}{3}x^2+\frac{1}{3}x+1$ since $(x+1)^3=x^3+3x^2+3x+1$. Prove that we can choose a positive integer n such that the sum of all coefficients of the resultant polynomial is less than 2.023.

Solution. Let us assume $n>1$, or otherwise the result is trivial. It is the same as proving

$$\binom{n}{0}^{-1}+\binom{n}{1}^{-1}+\cdots+\binom{n}{n}^{-1}<2.023$$

for some positive integer $n>1$. Indeed, for any $n\geqslant 4$ and $2\leqslant j\leqslant n-2$, we have $\binom{n}{j}\geqslant\binom{n}{2}=\frac{n(n-1)}{2}$. Therefore, we have

$$\sum_{j=0}^{n}\binom{n}{j}^{-1}=2\left(1+\frac{1}{n}\right)+\sum_{j=2}^{n-2}\binom{n}{j}^{-1}\leqslant 2+\frac{2}{n}+\sum_{j=2}^{n-2}\frac{2}{n(n-1)}=2+\frac{2}{n}+\frac{2(n-3)}{n(n-1)}.$$

Since both $\frac{2}{n}$ and $\frac{2(n-3)}{n(n-1)}$ tend to 0 when n tends to infinity, the above sum must be less than 2.023 for sufficiently large n. This proves the result. $\square$

Problem 12 (Bulgaria MO 2018 Problem 6)**.** There are x countries and y cities on a planet, where each country contains at least three cities (and every city belongs to a country). Some of the cities are connected by roads. For each city, it is connected to at least half of the cities in this country, and is connected to exactly one city in another country. For any two countries, there are at most two roads between cities of the two countries. Also, for any two countries with fewer than $2x$ cities in total, there is at least one road between cities of the two countries. Prove that there exists a cycle of at least $x+\frac{y}{2}$ cities.

Solution. We shall use Ore's theorem in the following solution, which states that a simple graph with $n\geqslant 3$ vertices is Hamiltonian if the sum of degrees of every pair of non-adjacent vertices is at least n.

Firstly, for each country, consider a simple graph where each vertex represents a city, and two vertices are adjacent if there is a road between the two cities. Suppose there are $k\geqslant 3$ vertices (cities) in total. By the given condition, the degree of each vertex is at least $\left\lceil\frac{k}{2}\right\rceil$. Therefore, the sum of degrees of every pair of vertices is at least $2\left\lceil\frac{k}{2}\right\rceil\geqslant k$. By Ore's theorem, there is a Hamiltonian cycle, i.e. a cycle passing through each city once.

Secondly, if $x=1$, we are done because we have already found a cycle of $y>1+\frac{y}{2}$ cities. If $x=2$, each city in each country is connected to a city in another country by a road. As there are at least 3 cities in each country, this gives at least 3 roads between the two countries, which is impossible. Therefore, we must have $x\geqslant 3$.

Consider the simple graph where each vertex represents a country, and two vertices are adjacent if there is at least one road between cities of the two countries. For any two non-adjacent vertices A and B, they must have at least $2x$ cities in total by the given condition. There is a road connecting each of these cities to another country different from A or B. Therefore, there are at least $2x$ roads connecting a city in A or B and a city not in A or B. Since there are at most two roads between cities of two countries, the roads correspond to at least $\frac{2x}{2}=x$ edges in the graph. In other words, the total degree of A and B is at least x, which is the number of vertices. By Ore's theorem, there is a Hamiltonian cycle, i.e. a cycle passing through each country once.

We can now form our cycle as follows. The cycle that we find above indicates the order of the countries that we travel, say $A_1,A_2,\ldots,A_x$. Each time when we enter a new country A_i at a city c (which is connected to a city in A_{i-1}), we would like to travel within the country to the city d such that d is adjacent to a city in A_{i+1} (which exists since there is an edge between A_i and A_{i+1}). Note that c and d must be distinct as each city is only connected to one city not in the same country. As there is a Hamiltonian cycle in A_i, we can always travel from c to d. Also, we can choose a direction of the cycle that passes through more number of cities. In this way, the number of roads that we have travelled is at least half of the number of cities in A_i.

This gives us one way to travel in a cycle. The number of roads that we have used is at least x (the number of vertices in the cycle of countries) plus half of the total number of cities, i.e. $x+\frac{y}{2}$. $\square$